Product Requirements Document

AiQi (AI 　) — Learn Claude Code

Version: 1.0 | **Date:** 2026-03-26 | **URL:** https://aiqi.pages.dev/

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1. Overview

AiQi is a bilingual (English/Chinese) tutorial app that teaches Claude Code to Chinese international graduates with zero coding experience. It takes learners from "never opened a terminal" to "shipped a live app" in 2 weeks through 14 interactive lessons.

Target audience: Chinese international graduates entering the job market who want to build with AI but have no coding background.

Core value proposition: 14 lessons. 2 weeks. Your first live app. No coding experience needed.

2. Tech Stack

Layer	Technology
Frontend	Vanilla TypeScript SPA (no framework)
Bundler	esbuild
Styling	CSS custom properties, mobile-first
Fonts	Inter (UI), Noto Sans SC (Chinese), JetBrains Mono (code)
Hosting	Cloudflare Pages
API	Cloudflare Pages Functions
Database	Cloudflare D1 (SQLite)
Auth	Google Identity Services
Analytics	Cloudflare Web Analytics
Runtime deps	0 (zero)
Dev deps	esbuild, typescript, wrangler

3. Design System

Colors

Token	Value	Usage
`--primary`	#6C5CE7	Buttons, links, active states
`--primary-light`	#A29BFE	Hover states, borders
`--primary-dark`	#5A4BD1	CTA gradients

`--accent`	#00B894	Success, completion
`--accent-light`	#55EFC4	Terminal output
`--warning`	#FDCB6E	Hints, paid badges
`--error`	#E17055	Error states
`--bg`	#FAFAFA	Page background
`--bg-terminal`	#1E1E2E	Terminal/code blocks

Typography

- Sans: Inter + Noto Sans SC (400/500/600/700)
- Mono: JetBrains Mono (400/500)
- Scale: 0.75rem to 2.5rem (8 steps)

Spacing

4px / 8px / 16px / 24px / 32px / 48px / 64px

Border Radius

8px / 12px / 16px / 24px / 9999px (pill)

4. Pages & Features

Route	View	Description
`#/`	Home/Landing	Hero with animated terminal mockup, feature cards, CTA
`#/lessons`	Lesson List	14 lessons in 2 weeks, with completion badges
`#/lesson/:id`	Lesson View	Step-by-step content with progress bar
`#/dashboard`	Progress Dashboard	Stats, streak, lesson map, continue button

Interactive Components

- **Terminal Simulator** — Black terminal UI with command input, typed output animation, fuzzy matching
- **Quiz Engine** — Multiple-choice with animated correct/incorrect feedback, retry without penalty
- **Copy-Paste Code Blocks** — One-tap copy with "Copied!" confirmation
- **Confetti Celebration** — CSS confetti + milestone modal on lesson completion

Bilingual System

- Instant EN/ZH toggle via header button
- `t()` function for UI strings (93 translation keys)
- Per-language lesson content JSON (28 files)
- Preserves scroll position on language switch

5. Component Architecture

```
app.ts          — Hash router, header, bottom nav, home/lesson list rendering
i18n.ts         — Language state, t() function, 93 translation keys
lessons.ts      — Lesson loader, step renderer, code blocks, navigation
terminal.ts     — Terminal UI, command matching, typing animation
quiz.ts         — Multiple-choice renderer, feedback, retry logic
auth.ts         — Google sign-in/out, localStorage persistence
progress.ts     — LocalStorage + API sync, streaks, dashboard rendering
celebrations.ts — Confetti particles, milestone modals
paywall.ts      — Freemium gate (currently disabled for testing)
```

Server-Side

```
functions/api/_shared.ts   — Google JWT verification, shared types, helpers
functions/api/auth.ts      — POST: verify Google token, upsert user
functions/api/progress.ts  — GET/POST: read/write lesson completion
```

6. Screenshots

AI起 AiQi

EN 中文 登录

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从零到上线
用你的语言

14节课。2周时间。你的第一个在线应用。
零基础也能学会。

```
$  claude "build me a personal website"
✓ Created index.html
✓ Added modern styling
🎉 Your site is live at mysite.pages.dev
```

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安全练习终端

首页 课程 进度

中英双语课程

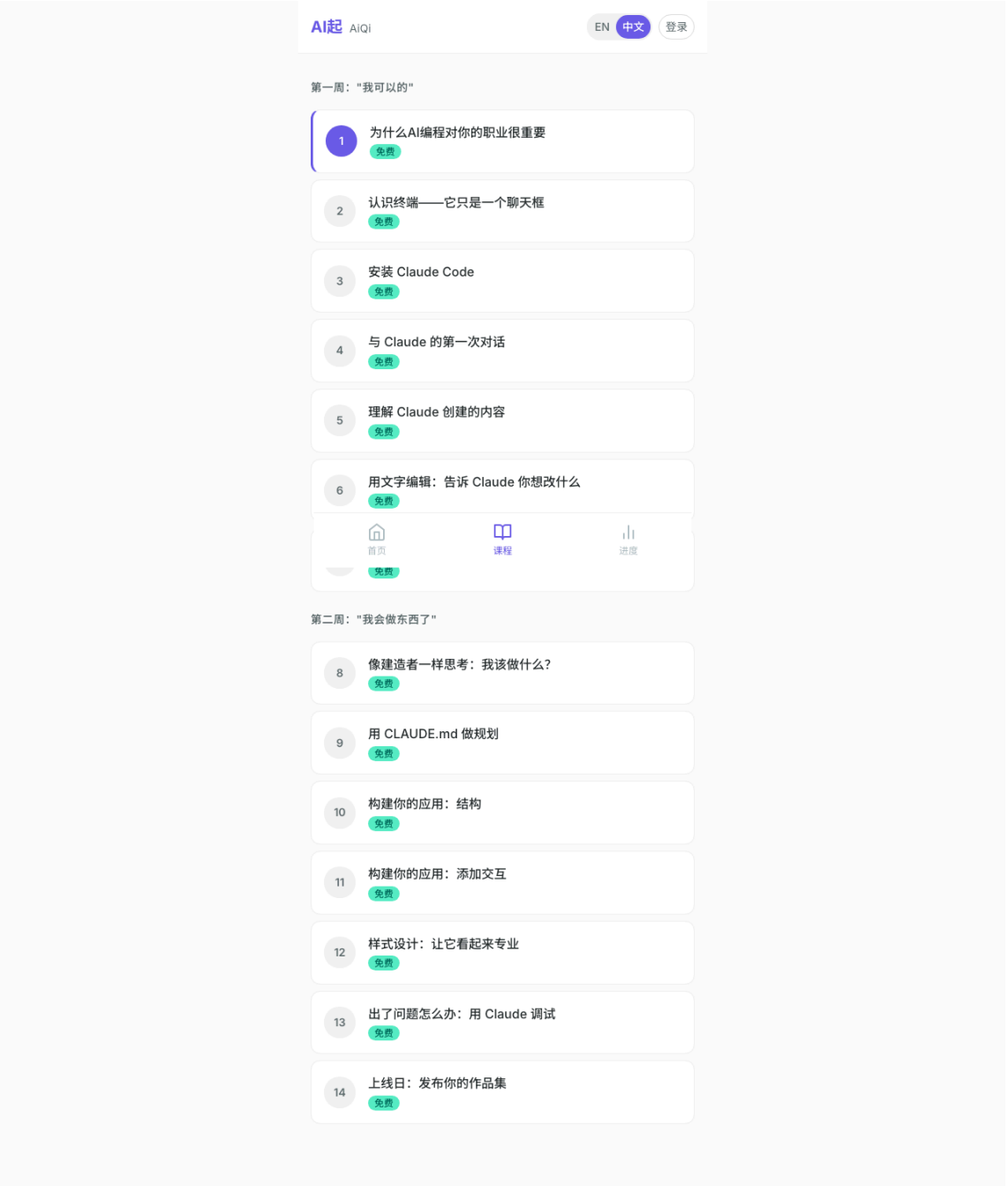
中文阅读，英文编程。随时切换。

循序渐进

每次只做一件事。绝不手忙脚乱。

上线真实项目

课程结束时你会有一个可以给雇主看的在线应用。



7. Accessibility

Feature	Implementation
Language	`<html lang="zh">`, dynamically updated
ARIA labels	Navigation, back button, terminal input, language toggle

ARIA roles	`role="progressbar"` on lesson progress, `role="status" aria-live="polite"` on quiz feedback
Navigation	`aria-current="page"` on active nav item, `aria-label="Main navigation"`
Keyboard	`:focus-visible` styles with 2px primary outline
Motion	`prefers-reduced-motion: reduce` disables all animations
Tap targets	Minimum 44px on all interactive elements (buttons use 48px)

8. Security

Measure	Implementation
JWT verification	Google tokeninfo endpoint verification on every API request
Audience check	Validates `aud` claim matches app's Google client ID
SQL injection	Parameterized D1 queries throughout
Input validation	lessonId validated as number 1-14 on server
XSS	innerHTML used only with author-controlled content (static JSON)
Secrets	`.env`, `.env.*`, `.dev.vars` in .gitignore
CORS	Same-origin (Cloudflare Pages serves both static + functions)

9. SEO

Tag	Value
Title	AiQi (AI) — Learn Claude Code
Description	Learn Claude Code with bilingual lessons. Zero coding experience needed — from terminal basics to shipping a live app in 2 weeks.
Canonical	https://aiqi.pages.dev/
OG Title	AiQi — Learn Claude Code in Your Language
OG Description	Bilingual tutorial app: from zero coding to shipping real projects with AI. 14 interactive lessons in English and Chinese.
OG Type	website
Twitter Card	summary
Robots.txt	Allow all, sitemap reference
Favicon	SVG rocket emoji

10. Deployment

```
# Build
npm run build          # esbuild → dist/

# Deploy
wrangler pages deploy dist/ --project-name=aiqi

# Type check
npx tsc --noEmit
```

Platform: Cloudflare Pages **Database:** Cloudflare D1 (aiqi-db) **Analytics:** Cloudflare Web Analytics **Repository:** https://github.com/Joshli316/AiQi

11. Open Items

Item	Priority	Status
Replace placeholder Google client ID	High	Blocked until Google OAuth credentials created
Payment integration (Stripe)	Medium	Deferred post-testing. Paywall code exists but is disabled
Unit tests for pure functions	Medium	getStreak(), parseHash(), getNextLesson()
GitHub Actions CI/CD	Low	Currently manual deploy via wrangler
Custom domain	Low	Using aiqi.pages.dev for now

Generated 2026-03-26. Source: https://github.com/Joshli316/AiQi